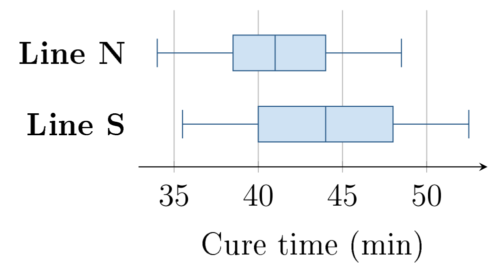
Two Test Plans for Cure Time

Unit 4 · Topic 4.9 · THE PROBLEM

Suppose line N uses a new formula for sealant, a material that closes gaps. Line S uses the standard formula. A team takes independent simple random samples of 18 of 600 recent N batches and 18 of 540 recent S batches. Mean cure times are 41.2 and 44.0 minutes; the boxplots are shown. Before sampling, the team asked, "Do the two populations differ in mean cure time?" Formula was not randomly assigned to lines.



Rhea: "Keep the original question and use population means. A difference between lines may have other causes."

Sol: "Since 41.2 < 44.0, use a lower tail with sample means. A significant result proves the formula caused faster curing."

FIRST TAKE — one sentence before you work the parts

Did the team originally ask whether N was faster or whether the lines differed in either direction? Explain in one sentence.

DOK 1 **(a)** Define μ_N and μ_S in context.

DOK 2 **(b)** State H_0 and H_a in N-minus-S order to match the question asked before sampling.

DOK 3 **(c)** In 3–4 sentences, judge the two plans. Explain why the original question sets the tail, why hypotheses use population means, and why these two lines cannot isolate the formula as the cause.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.